

Conv_Simple — Ignition Perspective Dashboard

Step-by-step: build the live web view in the Ignition Designer · 2026-05-29

What you're about to do — in plain English. You'll create a web page inside Ignition (a "Perspective view") that shows the conveyor's live status — its inputs, outputs, heartbeat, and (optionally) VFD readings. Anyone on your Tailscale network, including your phone, can then open it in a browser. The hard parts — talking to the PLC and reading its data — are *already done and verified live*. This guide is only about laying out the screen. Budget ~10–15 minutes.

Already set up for you (don't redo these)

Piece	Value	State
Ignition Gateway	<code>http://100.72.2.99:8088</code> (or <code>localhost:8088</code>)	running
PLC device	<code>MIRA_PLC</code> — Micro800 / CIP, <code>192.168.1.100:44818</code>	enabled, live
Tags	<code>[default]MIRA_IOCheck/</code> — DI_00–11, DO_00–07, heartbeat	defined
You provide	The Ignition Designer Launcher (desktop app)	open it

Steps

- 1

Open the Designer and a project. Launch *Ignition Designer Launcher* → add/select the gateway `http://100.72.2.99:8088` → **Open Designer** → log in. Open the `Mira_tags` project, or make a fresh one: *File* → *New Project*, name it `ConvSimpleLive` .

Why: Perspective views live inside a project. A new project keeps this dashboard isolated from the existing (and partly broken) `ConveyorMIRA` .
- 2

Confirm the tags are live. In the *Tag Browser* (bottom-left), expand `default` → `MIRA_IOCheck` . The DI/DO tags should show a **green** quality dot and real values that flip as you toggle inputs.

If they're red/stale: the PLC link is down — check the Ethernet cable to the PLC switch and that the PLC is powered and in RUN (`Get-NetAdapter Ethernet` = Up).

- 3 **(Optional) Add the VFD analog tags.** The existing tags are digital I/O only. To show DC bus / frequency, open the *OPC Browser* (Tag Browser → the OPC plug icon) → expand `Ignition OPC UA Server` → `[MIRA_PLC]` → drag `vfd_dc_bus`, `vfd_frequency`, `vfd_current`, `vfd_voltage`, `vfd_cmd_word` into a new folder `MIRA_IOCheck/VFD`.

These are raw integers. Scale them with the tag's *Engineering* settings or an expression tag: DC bus & voltage $\div 10$, frequency & current $\div 100$.

- 4 **Create the view.** In the *Project Browser*, right-click *Perspective* → *Views* → *New View*. Name it `ConvSimpleLive`, root container type **Flex Column**, tick "Make this the default page" if offered.
- 5 **Add a title.** From the *Components* palette drag a **Label** to the top. Set its `props.text` to `Conv_Simple – Live`.
- 6 **Add the value tiles (drag-to-bind).** The fastest way: drag a tag *straight from the Tag Browser onto the view* — Ignition offers to create a bound component. Do this for `VFD/vfd_dc_bus` (pick *Label* or *LED Display*), then `vfd_frequency` and `vfd_current`. Each lands pre-bound to its tag.

Why drag-to-bind: it writes the tag binding for you, so you can't mistype the path.

- 7 **Add status indicators.** Drag each boolean tag onto the view and choose an **LED Display**: `Inputs/DI_02` (E-stop NC), `Outputs/DO_00` (green light), `Outputs/DO_02` (contactor / drive), `Diagnostics/heartbeat`. Give each a Label beside it. (See the reference table for what each maps to.)
- 8 **Save.** `Ctrl + S`. Saving publishes the view to the gateway immediately.
- 9 **View it in a browser.** Open `http://100.72.2.99:8088/data/perspective/client/ConvSimpleLive` (swap the project name if different). You should see the live values updating.
- 10 **Open it on your phone.** With Tailscale on, browse to the same URL. That's the always-on web dashboard — the Ignition route, done.

For 24/7 without the laptop, the Ignition gateway must move to an always-on box on the PLC switch (the edge gateway in the design spec). Until then, keep the laptop on.

Tag reference (what to bind)

Purpose	Tag path	Component
E-stop NC channel (healthy = 1)	[default]MIRA_IOCheck/Inputs/DI_02	LED Display
Run pushbutton	[default]MIRA_IOCheck/Inputs/DI_04	LED Display
Green light	[default]MIRA_IOCheck/Outputs/DO_00	LED Display
Red light	[default]MIRA_IOCheck/Outputs/DO_01	LED Display
Contactor / drive enable	[default]MIRA_IOCheck/Outputs/DO_02	LED Display
Heartbeat (scan alive)	[default]MIRA_IOCheck/Diagnostics/heartbeat	LED Display
VFD DC bus ($\div 10 \rightarrow V$)	[default]MIRA_IOCheck/VFD/vfd_dc_bus	Label / Gauge
VFD frequency ($\div 100 \rightarrow Hz$)	[default]MIRA_IOCheck/VFD/vfd_frequency	Label / Gauge
VFD current ($\div 100 \rightarrow A$)	[default]MIRA_IOCheck/VFD/vfd_current	Label
VFD command word (1=STOP, 18=FWD)	[default]MIRA_IOCheck/VFD/vfd_cmd_word	Label

Full digital list available: `Inputs/DI_00 ... DI_11` and `Outputs/DO_00 ... DO_07` — bind whichever you need.

Troubleshooting

Symptom	Cause / fix
Tags red / "Bad_NotConnected"	PLC link down — check Ethernet into the PLC switch, PLC powered + in RUN, CIP port 44818 reachable.
View blank in browser	Wrong project name in the URL, or the view isn't saved. Re-check the <code>/client/<Project></code> path and press Ctrl+S.
VFD values look huge (e.g. 3200)	Raw integer not scaled — divide DC bus/voltage by 10, freq/current by 100.
Can't reach <code>:8088</code> from phone	Make sure Tailscale is connected on the phone and the gateway host is online on the tailnet.

Where this fits. This is the "proper SCADA" route. The lightweight MQTT route is already live at `http://100.68.120.99:8080`. Once both exist, Ignition can also publish its tags out via MQTT

(Sparkplug B) to the same broker — see the design spec [docs/superpowers/specs/2026-05-29-plc-web-dashboard-ignition-mqtt-design.md](#) .